ANUVAAD-ANALYTICS

AI-Powered Localized Public Grievance Management

Odisha Government E-Governance Framework

SYSTEM ARCHITECTURE & TECHNICAL WORKFLOW

Automated Translation, NLP Classification & Streamlit Analytics

The Governance Gap

PROBLEM STATEMENT

Language Barriers in Public Redressal

- Language Divide: 85%+ of citizens in rural Odisha write grievances ir
- Systemic Disconnect: Administrative classifications & reporting run ir
- Manual Redressal Bottlenecks: Physical translation is slow, expensiv
- No Priority Insight: Missing automatic sentiment/urgency evaluation c
- Siloed Operations: Lack of direct statistical mapping of district/depart

85%

Rural Grievance Volume

is submitted in native Odia, demanding automated local translation and processing.

Pipeline Architecture

WORKFLOW MODEL

The Automated Anuvaad Pipeline Flow:

1. Citizen Input

Grievance submitted in Odia text (e.g. BSKY/Kalia schemes)

2. Translator

Indicator API / Google Translate engine translates to English

3. AI Classifier

TF-IDF & Logistic Regression predicts target department

4. Sentiment Engine

TextBlob calculates polarity & assigns Urgency flags

5. DB & Dashboard

Logs to SQLite and renders metrics in Streamlit Dashboard

RESULT: Instant department routing, real-time visualization, and urgent negative complaint escalation.

Indic Translation Engine

ODIA-TO-ENGLISH CONVERSION

IndicTrans2 & deep-translator Bridge

- API Integration: Powered by deep-translator mapping Odia ('or') to E
- Accuracy: Handles complex Odia conjugations and administrative ph
- Sanitization: Removes text symbols, punctuation, and handles doubl
- Self-Healing Offline Fallback: Integrates local dictionary matching as
- Robustness: Ensures 100% service uptime even in low or offline net

Offline Local Dictionary

Matches Odia keys to Eng meanings:

- ■■■■■ ■■■■■ -> **Kalia scheme**
- ■■■■■■■■■ -> **BSKY health card**
- ■■■■■■■■■■ -> **Hospital**
- ■■■■■ ■■ -> **Drinking water**
- ■■■■ / ■■■■■■ -> **Police / FIR**
- ■■■■■■■■ ■■■■ -> **Mid-day meal**

AI Department Classifier

NLP ROUTING ENGINE

Automated Machine Learning Categorization

- Model Setup: TF-IDF Vectorizer coupled with a Logistic Regression c
- Automatic Training: Auto-trains on start if models are missing (self-he
- Odisha-Specific Focus: Tuned with localized vocabulary (BSKY, Kali
- High Efficiency: Executes predictions in milliseconds, bypassing hum
- Target Departments: Routes complaints to 6 major sectors (Agricultu ome, U

Classification Scope

1. Agriculture & Farmers
2. Health & Family Welfare
3. School & Mass Education
4. Panchayati Raj & Water
5. Home Department (Police)
6. Housing & Urban Development

Sentiment & Priority Analytics

URGENCY DETECTION

Harnessing Emotion for Prioritization

- Sentiment Lexicon: Powered by TextBlob analyzer evaluating polarity
- Smart Categorization: Automatically labels sentiment: Positive, Neutr
- Urgency Escalation: Negative sentiment instantly flags grievances as
- Prioritized Action: Critical complaints are moved to the top of the offic
- Empathy Mapping: Helps administration identify distressed areas and

Priority Mapping Rule

Polarity < -0.1 --> Negative (Critical)

Polarity > 0.1 --> Positive

Otherwise --> Neutral

CRITICAL ALERT SYSTEM

Severe grievances bypass regular queues, alerting officers immediately.

Officer Analytics Dashboard

ADMINISTRATIVE PANEL

E-Governance Monitoring Hub

- Executive KPIs: Live cards showing Total, Pending, Resolved, and R
- Interactive Charts: Plotly graphs displaying department loads and reg
- Geographical Heat Map: District-wise breakdown representing top gr
- Timeline Trends: 30-day inflow logs displaying historical filing velocity
- Control Center: Officers can filter, re-assign departments, update sta

Dashboard Core Metrics

Total Active cases	14 Active Cases
Resolution Rate	35.7% Closed
Critical Grievances	13 Flagged Cases

Impact & Value Proposition

KEY GOVERNANCE OUTCOMES

Language-Inclusive Governance

Empowers non-English speaking rural citizens to raise issues directly in Odia, making governance accessible to all.

Operational Automation

Reduces inter-departmental routing times from weeks to milliseconds, eliminating manual administrative delays.

Empathetic Prioritization

Automatically extracts emotional urgency from text, prioritizing distressed citizens in critical need.

Data-Driven Decisions

Renders visual analytics of district and department performance, highlighting bottlenecks for policy makers.